

P425/1
Pure Math
Paper 1
3 hours

UACE MOCK EXAMINATIONS 2008
P425/1 PURE MATHEMATICS
PAPER 1
TIME: 3 HOURS

Instructions:

Attempt **all** questions in section A, and **NOT** more than five from section B.

SECTION A (40 MARKS)

1. A curve passes through the point (2,0). The gradient of the normal to the curve at any point (x,y) is $\sec y - x \tan y$. Find the equation of the curve. (5 marks)
2. Prove by induction: $\sum_{r=1}^n r(r+1) = \frac{1}{3} n(n+1)(n+2)$ (5 marks)
3. Find the coordinates of the foot of the normal from the point (4,1, - 3) to the plane $x + 2y - z = 3$ (5 marks)
4. Find the different number of combinations of the letters of the word BOOKLET taking three letters at a time. (5 marks)
5. Show that $\int_{\pi/4}^{\pi/2} \operatorname{Cosec} 2\theta \cot \theta \, d\theta = \frac{1}{2}$ (5 marks)
6. Solve for x in the equation: $\tan^{-1} x + \sin^{-1} \frac{3}{5} = \frac{\pi}{4}$ (5 marks)
7. At the instant when the circumference of a circle is 5cm the radius s increasing at a rate of 0.5 cms^{-1} . If this rate remains constant find how long it takes to double the radius, and determine the rate at which the area of the circle is increasing at the end of this time. (5 marks)
8. Show that the circles $(x - 1)^2 + (y + 3)^2 = 64$ and $(x + 2)^2 + (y - 1)^2 = 9$ touch each other; and find the coordinates of their point of contact. (5 marks)

SECTION B (60 MARKS)

9. (a) Prove that $e^{x \ln x} = x^x$, hence or otherwise differentiate

(i) 10^{x^2}

(ii) $x^{\sin x}$

(6 marks)

- (b) A rectangular tank with a square base is to be built inside a right circular conical shelter of radius R and height H . Prove that the maximum volume of the tank is $\frac{8}{27}R^2H$

(6 marks)

10. (a) Find the component of the vector $5\mathbf{i} - 4\mathbf{j} + 8\mathbf{k}$ in the direction $2\mathbf{i} - \mathbf{j} + 2\mathbf{k}$

(b) Given that the angle between the line $\mathbf{r} = \begin{pmatrix} 5 \\ 0 \\ 4 \end{pmatrix} + t \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$

and the plane $3x + \lambda y + z = 10$ is 30° , find the values of λ . Calculate the acute angle between the planes corresponding to these values of λ

(12 marks)

11. (a) Solve for the complex number Z in the equation

$$iz + 2\bar{z} = 10 + 11i$$

- (b) Express $z = \sqrt{3} + i$ and z^3 in modulus – argument form and represent z and on the same Argand diagram clearly labeling the points representing them; hence deduce $|z||z - 1||z + 1|$.

(7 marks)

12. $A(x_1, y_1)$ and $B(x_2, y_2)$ are two points on the parabola $y^2 = 4ax$

- (a) Prove that the equation of the chord $\overline{AB}$ is $(y_1 + y_2)y - 4ax = y_1y_2$; and deduce the equations of the tangents at A and B

- (b) The tangents at A and B meet in point C ,

- (i) Show that the y – coordinatae at C is $y = \frac{2a}{M}$; where m is the gradient of $\overline{AB}$

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- (ii) Prove that the line joining B to the origin $(0,0)$ meets the tangent at A in point

$$\left(\frac{2ax_1x_2}{y_1y_2 - 2ax_2}, \frac{2axy_2}{y_1y_2 - 2ax_2} \right)$$

(12 marks)

13. Give the curve $y = \frac{2(x^2 - x - 2)}{x^2 + x - 12}$

(a) Find

(i) the value of x as $y \rightarrow \pm \infty$

(ii) the value of y as $x \rightarrow \pm \infty$

(b) Find the points at which the curve cuts the lines

(i) $x = 0$

(ii) $y = 0$

(iii) $y = 2$

(c) Sketch the curve

(12 marks)

14. (a) Solve the equation: $1 + \sin 2\theta + \cos 2\theta = \cos \theta + \sin \theta$; for $0^\circ \leq \theta \leq 360^\circ$

(b) Prove that, in any triangle ABC, $\frac{a^2}{b^2 - c^2} = \frac{\sin(B + C)}{\sin(B - C)}$; hence find the angles

A and B given that $C = 45^\circ$, $a = b\sqrt{2} = 2c$.

(12 marks)

15. A man's salary increases at a rate proportional to the product of the value of the salary and the time he takes working. His initial salary of 10 million shillings increases to 15 million in 2 years.

(a) Calculate the rate of which his salary is increasing at the end of 5 years.

(b) Given that the salary increases from 14 million to P million in 2 years, find the value of P

16. (a) Evaluate

$$\int_1^4 \left(\frac{4-x}{2+\sqrt{x}} \right)^2 dx$$

(5 marks)

(b) Show that $\int_0^{\pi/2} \frac{d\theta}{(e^{\sin \theta} + 1) \sec \theta} = \ln \left(\frac{2e}{e+1} \right)$

(7 marks)

END